[Date]

The Editor-in-Chief
IEEE Access

Dear Editor,

We submit the manuscript "Sensor-Driven Tilting Agrovoltaic Canopy for Sustainable Farming: Design, Simulation, and Prototype at TCE Madurai" for consideration as a research article in IEEE Access.

Agrovoltaic systems share land between crops and solar panels, but a fixed panel angle favours either the crop or the electricity. We define a co-optimization accuracy that scores a tilt strategy against the best possible angle for the same objective, and we present a six-sensor tilt controller that reaches 99.6 to 99.7 percent of that best score at seven climates in simulation, against 89.1 to 91.5 percent for a fixed 45 degree tilt. The paper also reports a simulated month driven by real NASA POWER weather and describes a wire-driven canopy and a scaled prototype.

Why this journal (our judgement, please check the journal's aims and scope): Multidisciplinary open-access engineering journal that publishes simulation-based engineering studies; the controller, the simulation and the implementation sections fit it.

What we state openly: all results are simulation and the sensors are software models of named commercial devices; the accuracy is measured against the controller's own objective, not against crop yield; no field measurement is reported. Reviewers usually ask for reproducible code and a comparison with existing control methods: the code and data are on GitHub and the paper compares against fixed, tracking and monthly-plan designs.

[Authors to confirm before sending: the work is original, has not been published and is not under consideration elsewhere; all authors have approved the manuscript; there is no conflict of interest, or state it.]

Code, data and results are available at https://github.com/thanwin19/agrovoltaic-canopy-srmc.

Suggested reviewers: [to be completed by the authors].

Sincerely,
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